FUNCTIONAL COEFFICIENT MODELS UNDER UNIT ROOT BEHAVIOR

TED JUHL

Abstract. We analyze the statistical properties of nonparametrically estimated functions in a functional-coefficient model if the data has a unit root. We show that the estimated function converges at a faster rate than under the stationary case. However, the estimator has a mixed normal distribution so that point-wise confidence intervals are calculated using the usual normal distribution theory rather than a Dickey-Fuller distribution. We illustrate the estimation procedure using U.S. unemployment and interest rate data.

1. Introduction

The analysis of time series models has recently moved toward modeling nonlinear features of the data. Some examples of these models include threshold autogressive models (TAR) as introduced by Tong (1978, 1990) and smooth transistion autoregressive (STAR) models beginning with Bacon and Watts (1971). The large sample theory for parameters estimated and tests associated with these models is analyzed in Chan (1991), Chan (1993), Chan and Tsay (1998), and Hansen (1996) among others. A procedure for analyzing STAR models was presented in Terasvirta (1994). Cai, Fan, and Yao (2000) recently provided a technique to analyze functional-coefficient models using local linear estimation techniques. The procedure uses a nonparametric smoothing method to estimate autoregressive coefficients that are potentially data dependent. The functional-coefficient model are shown to be a valuable generalization to TAR and STAR models, allowing for multiple regime modes and unknown functional forms.

Version: March 11, 2002. My thanks go to Bruce Hansen, Shigeru Iwata, Roger Koenker and Zhijie Xiao for suggestions that greatly improved the paper. Ted Juhl, Department of Economics, University of Kansas (juhl@ukans.edu).

When the data have unit roots, the analysis of nonlinear models often becomes nebulous, with both nonlinearity and unit roots often mistaken for the other. For example, even a simple break in a deterministic term can be mistaken for a unit root (Perron (1989)) or a unit root process may be appear to generate a "spurious break" (Nunes, Kuan, and Newbold (1995)). In the case of TAR models, Caner and Hansen (2001) develop the asymptotic theory associated with a two-regime TAR model when the data contains a unit root. They show that the presence of a unit root changes the limiting distributions of tests for threshold effects, and that tests for a unit root based on estimating threshold models are dependent on the threshold effect.

In this paper, we analyze the local linear functional coefficient estimation technique of Cai, Fan, and Yao (2000) when the true process is a nonstationary (linear) unit root process. The motivation for such a task is apparent. In simple linear regression, the presence of a unit root changes the asymptotic distribution of the autoregressive parameter estimator, as is well known from Dickey and Fuller (1979). Moreover, in the case of M-estimators, unit roots also give rise to nonstandard limiting distributions which change with the choice of M-estimator (see Cox and Llattas (1991) and Lucas (1995)). Many time series that are explored using nonlinear or nonparametric techniques are considered to be potentially nonstationary in the sense that a unit root is sometimes suspected. Hence, it is of interest to examine how these estimation methods are affected by the presence of a unit root.

There are several questions we address in the paper. First, we do not know in advance if the data has a unit root. If we proceed and estimate functional-coefficient models, do we get nonsense results? Does a unit root introduce "spurious nonlinearity" in functional-coefficient models? Next, is there the usual independence between the distribution of coefficients on the nonstationary terms and the stationary terms? Can we use the asymptotic results to find evidence for or against the unit root hypothesis? The answers to the above questions are no, no, no, and yes.

The structure of the paper is as follows. In Section 2, we review the estimation of functional-coefficient models from Cai, Fan, and Yao (2000). We find the asymptotic

distribution of the estimator under a pure unit root process in Section 3. In Section 4, we consider the more realistic case of higher order autoregressive processes. We provide a Monte Carlo experiment in Section 6 to assess the validity of the asymptotic results. Section 7 presents an empirical example using U.S. unemployment rates. Section 8 concludes.

Throughout the paper, $\xrightarrow{d}$ signifies convergence in distribution.

2. Estimation

Suppose that u_t, x_t, y_t are jointly strictly stationary and that the conditional mean is

$$E(y_t|x_t, u_t) = \sum_{j=1}^{p} a_j(u_t)x_{tj}. \tag{2.1}$$

The motivation for assuming the conditional mean of this form is to provide some structure relative to a completely nonparametric model. However, since there is no functional form imposed on the coefficients a_j, there is sufficient flexibility remaining to encompass a wide class of models, including the TAR and STAR models mentioned earlier. In addition, since the coefficients are an unspecified function of u_t, we allow for very general forms of TAR and STAR models, perhaps with several regimes. This allows for the estimation of a very large class of models of likely interest to researchers in time series econometrics. Moreover, by providing the limited structure, one potentially reduces the well known "curse of dimensionality" arising from a completely nonparametric setting.

There have been a variety of approaches for estimating functional-coefficients, such as Chen and Tsay (1993) and Hastie and Tibshirani (1993). The local linear nonparametric estimator of the functional coefficient model at u given in Cai, Fan, and Yao (2000) is defined as $\hat{a}_j(u)$ where $\hat{a}_j(u)$ minimizes the sum of weighted squares

$$\sum_{t=1}^{T}\left[y_t - \sum_{j=1}^{p}\{a_j + b_j(u_t - u)\}x_{tj}\right]^2 K_h(u_t - u). \tag{2.2}$$

$K_h(\cdot) = h^{-1}K(\cdot)$, h is a bandwidth, and $K(\cdot)$ is a kernel function.

Let $\alpha(u) = (a_1(u), a_2(u), \ldots, a_p(u), b_1(u), b_2(u), \ldots, b_p(u))^\top$. Then the solution to the minimization problem is given as

$$\hat{\alpha}(u) = H^{-1} R_T^{-1} S_T, \tag{2.3}$$

with

$$R_T = R_T(u) = \begin{pmatrix} R_{T,0} & R_{T,1} \\ R_{T,1} & R_{T,2} \end{pmatrix}, \quad S_T = S_T(u) = \begin{pmatrix} S_{T,0} \\ S_{T,1}, \end{pmatrix}$$

where

$$R_{T,i} = R_{T,i}(u) = \frac{1}{T} \sum_{t=1}^{T} x_t x_t^\top \left(\frac{u_t - u}{h} \right)^i K_h(u_t - u), \quad i = 0, 1, 2,$$

$$S_{T,i} = S_{T,i}(u) = \frac{1}{T} \sum_{t=1}^{T} x_t \left(\frac{u_t - u}{h} \right)^i K_h(u_t - u) y_t, \quad i = 0, 1 \quad \text{and,}$$

$$H = \begin{pmatrix} I_p & 0 \\ 0 & h \times I_p \end{pmatrix}$$

Cai, Fan, and Yao (2000) provide a statistical analysis of the above estimator under assumptions of stationarity and strong mixing data. They show that the estimator of $a(u)$ is asymptotically normally distributed with the usual nonparametric convergence rate of $\sqrt{Th}$.

3. Asymptotics for Unit Root Processes

In this section, we describe the asymptotic distribution of the nonparametric estimator of the functional coefficient model under unit root behavior. We suppose that the data generating process follows

$$y_t = y_{t-1} + \epsilon_t$$

There are many results for the above model when linear models are estimated. It is well known, (e.g. Dickey and Fuller (1979)) that when estimating the the coefficient on y_{t-1} in a time series model results in a non-standard limiting distribution and a convergence rate of T rather than $\sqrt{T}$.

Caner and Hansen (2001) estimate TAR models using $u_t = \Delta y_{t-r}$ or $u_t = y_{t-1} - y_{t-r}$ for some $m > 0$, respectively as the threshold variable. That is, the autoregressive parameters are allowed to take two sets of values depending on the magnitude of u_t.

We consider the asymptotic disbribution of $\hat{a}(u)$ under choices for u_t that include these possibilities. From the notation of the estimation section, we have $x_{t1} = y_{t-1}$ and we are estimating the autoregressive parameter (which is one) using $\hat{a}_1(u)$.

Since substantial structure is already imposed on the data generating process above, minimal assumptions are needed. We state the following theorem.

Theorem 3.1. *Suppose that the data is generated by $y_t = y_{t-1} + \epsilon_t$, where ϵ_t is an i.i.d. sequence with $E(\epsilon_t^2) = \sigma^2$, $E(\epsilon_t)^4 < \infty$. Let $\mathcal{F}_t = \sigma(\epsilon_t, \epsilon_{t-1}, \ldots)$ and suppose that u_t is $\mathcal{F}_{t-1}$ measurable. In addition suppose that $h \to 0$ and $Th^2 \to \infty$. $K(u)$ is a bounded symmetric density with $\lim_{|u|\to\infty} K(u) = 0$. Let $f(u)$ be the density of u and suppose that f has a continuous derivative. Then*

$$T\sqrt{h}(\hat{a}_1 - 1) \xrightarrow{d} MN(0, V_1) \tag{3.1}$$

where

$$V_1 = \frac{\int K(x)^2 dx}{f(u)} \left(\int W_1(r)^2 dr \right)^{-1},$$

$W_1(r)$ is a standard Brownian motion, and $MN(0, V_1)$ denotes a mixed normal distribution.

and $u_t = y_{t-1} - y_{t-m}$ or $u_t = \Delta y_{t-m}$ for some $m > 0$.

Remark 1: Simple examples of u_t satisfying $\mathcal{F}_{t-1}$ measurability include standard choices of threshold variables in TAR models such as $u_t = y_{t-1} - y_{t-r}$ or $u_t = \Delta y_{t-r}$ for some $r > 0$.

Remark 2: The limiting distribution of the coefficient on y_{t-1} converges at rate $T\sqrt{h}$. The coefficient from linear regression converges at rate T so this is an expected result. However, the linear regression coefficient has a Dickey-Fuller distribution whereas the functional-coefficient estimator is mixed normal. It is easy to see that the estimated asymptotic variance will converge to the conditional variance of the resulting mixed normal distribution. This implies that a t-ratio will have a standard normal distribution. We will explore this result in the Monte Carlo section as well.

Remark 3: In the proof of the theorem, it is apparent that if h is constant, the limiting distribution is a linear combination of a Dickey-Fuller distribution and the

mixed-Normal distribution. A similar distribution arises in unit root tests using stationary covariates in Hansen (1995), or in unit root tests with ARCH in Seo (1999). As $h \to 0$, the weight on the Dickey Fuller distribution goes to 0. If $h \to \infty$, we return to linear regression, and the weight on the Dickey Fuller distribution is one.

Remark 4: Phillips and Park (1998) show that if one performs standard nonparametric regression (Nadaraya-Watson) of y_t on y_{t-1} in a pure unit root process, the estimator also has a mixed normal distribution. However, the convergence rate of the estimator is at most $T^{\frac{1}{4}}$.

Up to this point, we have assumed that the process (4.1) has no constant. Suppose we consider

$$y_t = \mu + \rho y_{t-1} + +\epsilon_t \tag{3.2}$$

Then if a researcher demeans the data before estimating the a functional-coefficient model, then the right hand side variable becomes $y_t - \bar{y}$ and the first left hand side variable becomes $y_{t-1} - \bar{y}_{-1}$. We have the following result.

Corollary 3.1. *Given the assumptions from Theorem 3.1,*

$$T\sqrt{h}(\hat{a}_1 - 1) \xrightarrow{d} MN(0, V_2) \tag{3.3}$$

where

$$V_2 = \frac{\int K(x)^2 dx}{f(u)} \left(\int \left(W(r) - \bar{W} \right)^2 dr \right)^{-1},$$

and $\bar{W} = \int W(r)dr$.

Once again, the t-ratio will obtain asymptotic normality, and the construction of confidence intervals for the coefficient does not change if there is a unit root.

4. Higher Order Processes

A pure unit root process such as those we have discussed to this point are considered too simple for practical use. Higher order autoregressive processes provide a more

general data generating process to analyze economic time series. Suppose

$$y_t = \rho y_{t-1} + \sum_{k=0}^{q-1} c_k \Delta y_{t-1-k} + \epsilon_t. \tag{4.1}$$

Suppose that $\rho = 1$ and the roots of $c(z) = 1 - \sum_{k=0}^{q-1} c_k z^{k+1}$ lie outside the unit circle so that y_t has one unit root. The true coefficients of the functional-coefficient model are then $a_1 = 1, a_2 = c_0, \ldots, a_p = c_{q-1}$. Define the scaling matrix

$$D_{1T} = \begin{pmatrix} T^{-\frac{1}{2}} & 0_{1\times q} \\ 0_{q\times 1} & I_q \end{pmatrix}.$$

We have the following theorem.

Theorem 4.1. *Suppose that the data is generated by* 4.1 *and that the remaining conditions of Theorem 3.1 hold. Define* $\Delta Y_t = (\Delta y_{t-1}, \Delta y_{t-2}, \ldots, \Delta y_{t-q})^\top$ *and let* $\lambda = 1/c(1)$. *Then*

$$\sqrt{Th} D_{1T}^{-1}(\hat{a}(u) - a(u)) \xrightarrow{d} MN(0, V_3)$$

where

$$V_3(u) = \sigma^2 \frac{\int K(x)^2 dx}{f(u)} \begin{pmatrix} \sigma^2\lambda^2 \int W_1(r)^2 dr & \sigma\lambda E(\Delta Y_t^\top | u_t = u) \int W_1(r) dr \\ \sigma\lambda E(\Delta Y_t | u_t = u) \int W_1(r) dr & E(\Delta Y_t \Delta Y_t^\top | u_t = u) \end{pmatrix}$$

Remark 5: The distribution of the coefficients on the lagged differenced terms is *not* normally distributed as in the usual stationary case. Instead, all of the coefficients have a mixed normal distribution. Moreover, the covariance between the nonstationary coefficients and the stationary coefficients is nonzero.

Remark 6: There are additional differences between our results and those for the stationary case as in Cai, Fan, and Yao (2000). There is no bias term here since $a''(u) = 0$ for these coefficients. Second, since the error variance is constant, the variance "sandwich" disappears.

Similarly, it is easy to extend our results to demeaned data as in Corrollary 3.1. All of our results suggest that there is no change in the standardized distribution when we move from a stationary autoregressive process to a nonstationary process with a

unit root since confidence intervals will be calculated using the normal distribution. We explore this phenomenon in the next section via simulations.

5. Joint Confidence Intervals

So far we have analyzed the asymptotic distribution of the functional coefficient model at one point u. In practice, we probably want to estimate the function over a range of points rather than at a single point. In this section, we consider the joint distribution of the estimated functional-coefficients when our range contains a finite number of points.

Suppose that we are interested in estimating the functional-coefficient model over the points $(u_1, u_2, \dots u_M)$. Let $a(u_m)$ represent the $p = q+1$ coefficients in the model evaluated at the point u_m, $m = 1, \dots, M$. We stack the $a(u_m)$ vectors to get an $M \times p$ vector. The joint distribution of the estimated coefficients is given below.

Theorem 5.1. *Suppose that the data is generated by* 4.1 *and that the conditions of Theorem 3.1 hold.*

$$\sqrt{Th}(I_M \otimes D_{1T})^{-1} \begin{pmatrix} \hat{a}(u_1) - a(u_1) \\ \hat{a}(u_2) - a(u_2) \\ \vdots \\ \hat{a}(u_M) - a(u_M) \end{pmatrix} \xrightarrow{d} MN(0, V_4)$$

where

$$V_4 = \begin{pmatrix} V_3(u_1) & 0 & 0 & 0 \\ 0 & V_3(u_2) & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & V_3(u_M) \end{pmatrix}.$$

The structure of the covariance matrix suggests that there are no correlations between the estimated coefficients at different points u_m. However, in order to find joint confidence intervals for the coefficients at our M points, we need to adjust the width of the confidence bands. The Bonferroni method uses the fact that $P(\bigcap_{i=1}^{M} A_i) \geq 1 - \sum_{i=1}^{M} P(A_i^c)$. Hence, when constructing $(1-\alpha)$ confidence intervals,

we use a critical value associated with level $(1-\alpha/M)$ at each point so that we have a conservative confidence band around our estimated function. For comparison purposes, it is interesting to include the point-wise confidence intervals as an example of liberal confidence intervals. We illustrate the procedure in a later section.

6. Simulations

The functional-coefficient regression model is examined using linear models in this section. Our goal is to assess the accuracy of the normal approximation to the limiting distribution in the unit root case. To do this, we also include two stationary cases for comparison purposes.

The data is generated according to

$$y_t = \rho y_{t-1} + c_0 \Delta y_{t-1} + \epsilon_t,$$

where ϵ_t are i.i.d standard Normal. We consider $\rho = 0.85, 0.95$, and 1, and $c_0 = 0$ and 0.5. Samples of size 550 were generated and the first 50 observations were eliminated to mitigate starting effects.

We estimate functional-coefficient models using $u_t = y_{t-1} - y_{t-10}$. The bandwidth parameter takes values $h = cT^{\frac{-1}{5}}$ where $c = 0.5, 1$, and 2. The coefficient $\hat{a}(u)$ is calculated for values of u ranging from -3 to 3 by 0.05 as well as estimates of the asymptotic standard errors. Note that the calculation of standard errors is the same regardless of whether there is a unit root. We then construct a "t-ratio" at each point u for each replication. After performing 2000 replications, we then compare the resulting empirical distribution of the t-ratio to various quantiles of the standard Normal distribution. In particular, we consider $0.025, 0.05, 0.95$, and 0.975. That is, at each u, we calculate the percentage of replications where the statistic is less than -1.96 and -1.645, and we calculate percentage of replications where the statistic is greater than 1.645 and 1.96. The experiment is summarized in Figure 1 and Figure 2 for $c_0 = 0$.

We note that the asymptotic theory provides a very good approximation to the finite sample distribution for all bandwidths and values of ρ. Figure 1 essentially shows

the size of a one-sided test of the parameter value at each point in u at levels 2.5% and 5%. There appears to be no change in distribution as $\rho = 1$, just as suggested by the asymptotic results in the previous section. However, as the bandwidth increases, the finite sample distribution becomes less accurate for $\rho = 1$. The intuition for this effect is clear; as h increases, the limiting distribution involves a positive weight on the Dickey-Fuller distribution, a weight that reaches one when h is infinite. The Dickey-Fuller distribution itself is shifted to the left of a standard Normal with a 5% quantile at approximately -1.95 and a 95% quantile at 1.28. Figure 2 provides similar evidence.

When $c_0 = 0.5$, the limiting distribution again provides a reasonably accurate approximation of the empirical distribution. All of the above conclusions are similar. However, it appears that a larger bandwidth performs better when the coefficients on the lagged differenced terms are estimated. Figures 3 and 4 provide visual evidence of the performance of the approximation.

7. Empirical Examples

In this section, we estimate functional-coefficient models using two different U.S. time series. In both cases, we include the conservative Bonferroni bands when the function is estimated at 25 points (M=25). The point-wise confidence intervals are included for a liberal set of confidence bands.

7.1 Unemployment Rates

Caner and Hansen (2001) note that standard linear models suggest that the U.S. unepmoloyment rate contains a unit root. However, a unit root in unemployment rates is not plausible since the rates are bounded.

Several authors have estimated TAR models, STAR models or some variant thereof to examine the unemployment time series. Examples include Montegomery, Zarnowitz, Tsay, and Tiao (1998), Caner and Hansen (2001), Hansen (1997), and Chan and Tsay (1998), among others. We use the data from Caner and Hansen (2001) as well as their

specification of the variable that the coefficients depend on in the functional-coefficient model.

Consider the monthly unemployment rate for the U.S. from January 1956 to August 1999. Let y_t be the unemployment rate at time t. We define $u_t = y_{t-1} - y_{t-10}$ and we include 12 of the lagged differenced terms in our specification. We estimate the functional-coefficient model over a range which contains approximately 95% of the observations of u_t, which is from -1.3 to 2, and we divide this range into 25 equally spaced points. We use the differece Δy_t as the left hand side variable and y_t is demeaned. Bandwidths from $3\sigma_u T^{-.2}$ to $10\sigma_u T^{-.2}$ were considered.[1] Our main concern is the coefficient of $y_{t-1} - \bar{y}_{-1}$ since this coefficient is key for understanding the potential nonstationarity of unemployment. Since our left hand side variable is Δy_t, a unit root implies that $a_1 = 0$.

The results for all show very similar patterns and we provide the case where $h = 4T^{-.2}$ in Figure 3. We include a 95% point-wise confidence interval at each point as well as the conservative Bonferroni bands. From our initital analysis, we find some evidence against the unit root hypothesis. The pointwise confidence bands in the figure suggest that when the unemployment rate has increased by between 1 to 2 points over the previous nine month period, the autoregressive coefficient becomes closer to zero (a unit root). However, only 9 percent of our observations lie in this range, suggesting that the apparent unit root behavior may be the result of a small range of observations.

7.2 Interest Rates

We repeat our estimation procedure using three month Treasury rates over the same time period (1956 to 1999). Now y_t represents the three month Treasury rate and $u_t = \Delta y_{t-2}$. Again, we consider the model

$$\Delta y_t = a_1(u_t)(y_{t-1} - \bar{y}_{-1}) + \sum_{j=0}^{11} a_{j+1}(u_t)\Delta y_{t-1-j} + \epsilon_t.$$

[1] The estimator could not be computed for values much less than $2T^{-.2}$.

We analyze the model over a range containing 95% of the data and we divide the range into 25 equally spaced points. We estimate the model for bandwidths $3\sigma_u T^{-.2}$ to $10\sigma_u T^{-.2}$. The estimated coefficient $\hat{a}_1(u)$ when $h = 8\sigma_u T^{-.2}$ along with 95% point-wise confidence bands and the conservative Bonferroni bands are shown in Figure 6. The estimated functional-coefficient model has a differentr shape to the unemployment case. However, the unit root behavior appears to manifest itself over the portion of the data where u_t is negative.

Both of the above examples give evidence that a more general model may be useful for examining time series properties. In particular, using the functional-coefficient models and confidence bands, we find evidence against the unit root hypothesis.

8. Conclusion

We derive the limiting distribution of functional-coefficient models estimated using the local linear nonparametric regression technique developed in Cai, Fan, and Yao (2000) when the data contains a unit root. We show that if the bandwidth goes to zero at the prescribed rate, the limiting distribution is not Normal or a Dickey-Fuller distribution, but mixed normal. This implies that the point-wise t-ratio has a Normal distribution *regardless* of whether there is a unit root.

From Remark 2, we also see that if the bandwidth parameter used in the nonparametric regression is fixed, the limiting distribution is a linear combination of a Dickey-Fuller distribution and a Mixed Normal distribution. Hence, for a fixed bandwidth, the t-ratio does not have a Normal distribution. There is a familiar tradeoff between bias and variance when choosing a bandwidth parameter. This paper suggests that if one is concerned about nonstationarity in the form of unit roots, another factor enters the tradeoff. That is, for large values of the bandwidth, the distribution of the standardized estimated functional-coefficients also changes.

The estimation procedure is applied to unemployment rates and interest rates. For both sets of data, the unit root behavior is only apparent over part of the range of u.

Overall, the results in this paper suggest that funcitonal-coefficient models allow for a flexible treatment of autoregressive processes. Moreover, we find that there is no distributional change when moving from a stationary (linear) case to a unit root model. That is, the asymptotic theory indicates the usual "knife edge" results hold only for the rate of convergence and not for the form of the distribution under unit root behavior when functional-coefficient estimation is employed. In addition, the procedure introduced in the paper to find conservative confidence bands has the appealing feature that it does not require us to formulate a model for the coefficient functional form. Hence, the procedure allows us to detect such general forms of nonlinearity such as multiple regime TAR and STAR models without specifying any such parametric forms.

There are several topics left for future research. First, the choice of bandwidth could be treated using some sort of cross validation approach. In addition, it is possible to extend the results to show that the distributional results hold uniformly over u (infinite number of points). With this result in hand, tighter confidence bands may be possible using methods such as Härdle (1989).

Appendix A

The data generating process (4.1) with $\rho = 1$ is given by $y_t = y_{t-1} + \sum_{k=0}^{q-1} c_k \Delta y_{t-1-k} + \epsilon_t$. Let K_t denote $K((u_t - u)/h)$.

Lemma A.1.

$$D_T R_T D_T \xrightarrow{d} \begin{pmatrix} f(u) & 0 \\ 0 & f(u)\int x^2 K(x)^2 dx \end{pmatrix} \otimes A \tag{A.1}$$

where

$$D_T = I_2 \otimes \begin{pmatrix} T^{-\frac{1}{2}} & 0 \\ 0 & I_q \end{pmatrix},$$

and

$$A = \begin{pmatrix} \sigma^2\lambda^2 \int W_1(r)^2 dr & \sigma\lambda E(\Delta Y_t^\top | u_t = u) \int W_1(r) dr \\ \sigma\lambda E(\Delta Y_t | u_t = u) \int W_1(r) dr & E(\Delta Y_t \Delta Y_t^\top | u_t = u) \end{pmatrix}$$

Proof: First, consider

$$\frac{1}{T^2 h} \sum_{t=1}^{T} y_{t-1}^2 K_t.$$

By Theorem 3.3 of Hansen (1992),

$$\frac{1}{T^2} \sum_{t=1}^{T} y_{t-1}^2 \left(\frac{K_t}{h} - \frac{E(K_t)}{h} \right) \xrightarrow{p} 0.$$

Note that $\frac{1}{T^2} \sum_{t=1}^{T} y_{t-1}^2 \xrightarrow{d} \sigma^2\lambda^2 \int W_1(r)^2 dr$ (see Phillips and Solo (1992)) and

$$\begin{aligned} E\left(\frac{K_t}{h}\right) &= \int \frac{1}{h} K\left(\frac{u_t - u}{h}\right) f(u_t) du_t \\ &= \int K(x) f(xh + u) dx \to f(u) \end{aligned}$$

as $h \to 0$, giving the result for the first entry. Now consider

$$\frac{1}{T^2 h} \sum_{t=1}^{T} \left(\frac{u_t - u}{h}\right) y_{t-1}^2 K_t. \tag{A.2}$$

Since

$$\begin{aligned} E\left(\frac{(u_t - u)K_t}{h^2}\right) &= \int \left(\frac{u_t - u}{h^2}\right) K\left(\frac{u_t - u}{h}\right) f(u_t) du_t \\ &= \int x K(x) f(xh + u) dx \to f(u) \int x K(x) dx = 0, \end{aligned}$$

since $K(x)$ is a symmetric density so that (A.2) converges to zero in probability. The proof of

$$\frac{1}{T^2 h} \sum_{t=1}^{T} \left(\frac{u_t - u}{h}\right)^2 y_{t-1}^2 K_t \xrightarrow{d} f(u) \left(\int x^2 K(x)^2 dx \right) \sigma^2\lambda^2 \int W_1^2(r) dr$$

uses

$$\begin{aligned} E\left(\frac{(u_t - u)^2 K_t}{h^3}\right) &= \int \left(\frac{(u_t - u)^2}{h^3}\right) K\left(\frac{u_t - u}{h}\right) f(u_t) du_t \\ &= \int x^2 K(x) f(xh + u) dx \to f(u) \int x^2 K(x) dx. \end{aligned}$$

Now, again using Hansen (1992), we have

$$\begin{aligned}
T^{-\frac{3}{2}}h^{-1}\sum_{t=1}^{T} y_{t-1}\Delta Y_t K_t &= \frac{1}{T}\sum_{t=1}^{T}\frac{y_{t-1}}{\sqrt{T}}\Delta Y_t\frac{K_t}{h} \\
&= \frac{1}{T}\sum_{t=1}^{T}\frac{y_{t-1}}{\sqrt{T}}E\left(\Delta Y_t\frac{K_t}{h}\right)+o_p(1) \\
&= \sigma\lambda\int W_1(r)drE\left(\Delta Y_t\frac{K_t}{h}\right)+o_p(1).
\end{aligned}$$

Then

$$\begin{aligned}
E(\Delta Y_t K_t/h) &= \int E(\Delta Y_t|u_t)\frac{1}{h}K\left(\frac{u_t-u}{h}\right)f(u_t)du_t \\
&= \int E(\Delta Y_t|u_t = xh+u)K(x)f(xh+u)dx \\
&\to E(\Delta Y_t|u_t=u)f(u),
\end{aligned}$$

so that $T^{-\frac{3}{2}}h^{-1}\sum_{t=1}^{T} y_{t-1}\Delta Y_t K_t \xrightarrow{d} f(u)\sigma\lambda E(\Delta Y_t|u_t=u)\int W_1(r)dr$. The remaining terms are shown using similar arguments. The convergence results hold jointly so that (A.1) holds.

Lemma A.2. *Consider a point u. Let $Z_{T,t}$ be defined as*

$$Z_{T,t} = \begin{pmatrix} Z_{T,t,1} \\ Z_{T,t,2} \\ Z_{T,t,3} \\ Z_{T,t,4} \\ Z_{T,t,5} \end{pmatrix} = \begin{pmatrix} \frac{\epsilon_t}{\sqrt{T}} \\ \frac{\epsilon_t(K_t - E(K_t))}{\sqrt{Th}} \\ \frac{\epsilon_t\Delta Y_t(K_t - E(K_t))}{\sqrt{Th}} \\ \frac{\epsilon_t\left(\frac{u_t-u}{h}\right)K_t}{\sqrt{Th}} \\ \frac{\epsilon_t\Delta Y_t\left(\frac{u_t-u}{h}\right)K_t}{\sqrt{Th}} \end{pmatrix}$$

Then under the conditions of Theorem 3.1,

$$\sum_{t=1}^{[Tr]} Z_{T,t} \xrightarrow{d} B(r) = \begin{pmatrix} B_1(r) \\ B_2(r) \\ B_3(r) \\ B_4(r) \\ B_5(r) \end{pmatrix},$$

where $B(r)$ is a Brownian motion with covariance matrix

$$\Omega = \begin{pmatrix} \sigma^2 & 0 \\ 0 & f(u)\sigma^2 C \end{pmatrix}$$

where

$$C = \begin{pmatrix} \int K(x)^2 dx & \int xK(x)^2 dx \\ \int xK(x)^2 dx & \int x^2 K(x)^2 dx \end{pmatrix} \otimes \begin{pmatrix} 1 & E(\Delta Y_t^\top | u_t = u) \\ E(\Delta Y_t | u_t = u) & E(\Delta Y_t \Delta Y_t^\top | u_t = u) \end{pmatrix}$$

Proof: We show that the conditions of Theorem 4.1 of Hall and Heyde (1980) hold so that an invariance principle applies to the partial sums of $Z_{T,t}$. Consider the second element of $Z_{T,t}$. Let $V_t = \epsilon_t(K_t - E(K_t))$ and let $\mathcal{F}_t = \sigma(\epsilon_t, \epsilon_{t-1}, \ldots)$. Then V_t is a martingale difference sequence with respect to the filtration $\mathcal{F}_t$. We must show $E(\sum_{t=1}^T V_t^2/Th - E(\sum_{t=1}^T V_t^2/Th))^2 \to 0$ and $\sum_{t=1}^T E((V_t^2/Th)1(|V_t/\sqrt{Th}| > \eta)) \to 0$ for all $\eta > 0$. First,

$$E\left(\frac{1}{Th}\sum_{t=1}^T V_t^2 - \frac{1}{Th}\sum_{t=1}^T E(V_t^2)\right)^2 = E\left(\frac{1}{T^2h^2}\sum_{t=1}^T V_t^4 + \frac{1}{T^2h^2}\sum\sum_{t\neq s} V_t^2 V_s^2 - (EV_t^2/h)^2\right)$$

The first term is $O((Th)^{-1})$ since ϵ_t has fourth moments and $E(K_t^4) = O(h)$. The vector $(\epsilon_t, \Delta Y_t^\top, u_t)^\top$ is absolutely regular since each component is either i.i.d. or an AR process with innovations possessing an absolutely continuous distribution (see Mokkadem (1988)). Following Fan and Li (1999) and Li (1999), we can apply Lemma 1 of Yoshihara (1976) to obtain

$$\left|E\left(V_t^2 V_s^2\right) - E\left(V_t^2\right) E\left(V_s^2\right)\right| \leq M\beta(t-s),$$

where $M < \infty$ and $\sum_{j=0}^{\infty} \beta(j) < \infty$. Then

$$\left| E\left(\frac{1}{T^2 h^2} \sum \sum_{t \neq s} V_t^2 V_s^2 - (E V_t^2 / h)^2 \right) \right| = O((Th^2)^{-1}) = o(1).$$

Next,

$$\sum_{t=1}^{T} E\left(\frac{V_t^2}{Th} 1\left(\left| \frac{V_t}{\sqrt{Th}} \right| > \eta \right) \right) \leq \sum_{t=1}^{T} E\left(\frac{V_t^4}{T^2 h^2} \right) = O((Th)^{-1}).$$

The proofs for the remaining elements of $Z_{T,t}$ are similar. Joint convergence holds by using a multivariate invariance principle such as Theorem 27.17 of Davidson (1994). Calculation of the covariance matrix follows the proof of Lemma A.1.

Lemma A.3.

$$\sum_{t=1}^{T} \begin{pmatrix} \frac{y_{t-1}}{\sqrt{T}} \\ 1 \end{pmatrix} \otimes \begin{pmatrix} Z_{T,t,2} \\ Z_{T,t,3} \\ Z_{T,t,4} \\ Z_{T,t,5} \end{pmatrix} \Rightarrow \int F_1 \otimes dF_2 \tag{A.3}$$

where

$$F_1(r) = \begin{pmatrix} B_1(r) \\ 1 \end{pmatrix} \text{ and } F_2(r) = \begin{pmatrix} B_2(r) \\ B_3(r) \\ B_4(r) \\ B_5(r) \end{pmatrix}.$$

Since $B_1(r)$ and the remaining elements of $B(r)$ are independent, $\int F_1 \otimes dF_2$ has a Mixed Normal distribution with conditional covariance matrix $\int F_1 F_1^\top \otimes f(u)\sigma^2 C$ where C is defined in Lemma A.2.

Proof: From Lemma A.2, the of the first vector in the sum to $F_1(r)$. $Z_{T,t}$ is a martingale difference with respect to $\mathcal{F}_t$ allowing the application of Kurtz and Protter (1991) to (A.3) (see also Theorem 2.1 in Hansen (1992)).

Proof of Theorem 4.1: Let $x_t = (y_{t-1}, \Delta y_{t-1}, \ldots, \Delta y_{t-q})^\top = (y_{t-1}, \Delta Y_t^\top)^\top$. In addition, let

$$\tilde{x}_t = \begin{pmatrix} x_t \\ x_t \left(\dfrac{u_t - u}{h} \right) \end{pmatrix}.$$

Then

$$\begin{aligned} \sqrt{Th} D_{1T}^{-1} \left(\hat{a}(u) - a(u) \right) &= \sqrt{Th} (D_{1T} R_{T,0} D_{1T} - D_{1T} R_{T,1} R_{T,2}^{-1} R_{T,1} D_{1T})^{-1} \\ &\quad \times D_{1T} \begin{pmatrix} I_{q+1} & \vdots & -R_{T,1} R_{T,2}^{-1} \end{pmatrix} S_{T,\epsilon} \end{aligned}$$

where

$$S_{T,\epsilon} = \frac{1}{T} \sum_{t=1}^{T} \tilde{x}_t \epsilon_t K_t$$

We can write

$$\begin{aligned} \sqrt{Th} D_T S_{T,\epsilon} &= D_T \sum_{t=1}^{T} \begin{pmatrix} y_{t-1} Z_{T,t,2} \\ Z_{T,t,3} \\ y_{t-1} Z_{4,T,t} \\ Z_{T,t,5} \end{pmatrix} + D_T \sum_{t=1}^{T} \begin{pmatrix} y_{t-1} \epsilon_t E(K_t) / \sqrt{Th} \\ \epsilon_t \Delta Y_t E(K_t) / \sqrt{Th} \\ 0 \\ 0 \end{pmatrix} \\ &= G_{1,T} + G_{2,T} \end{aligned}$$

From the proof of Lemma A.1, $E(K_t) = O(h)$, so that $G_{2,T} = O(\sqrt{h})$. Now apply Lemma A.2 to show that $G_{1,T} \xrightarrow{d} G_1$ where G_1 has a Mixed Normal distribution with conditional covariance matrix

$$\begin{pmatrix} \int K(x)^2 dx & \int x K(x)^2 dx \\ \int x K(x)^2 dx & \int x^2 K(x)^2 dx \end{pmatrix} \otimes \sigma^2 f(u) A$$

with A defined in Lemma A.1. Therefore, using Lemmas A.1, A.2, and A.3,

$$\sqrt{Th} D_{1T}^{-1} \left(\hat{a}(u) - a(u) \right) \Rightarrow MN(0, V_3).$$

Proof of Theorem 5.1: The proof is similar to the proof of Theorem 4.1 but uses the fact that

$$E \left(\frac{1}{h} K \left(\frac{u_t - u_1}{h} \right) K \left(\frac{u_t - u_2}{h} \right) \right) \to 0$$

as $h \to 0$ for $u_1 \neq u_2$ so that the covariance matrix is block diagonal.

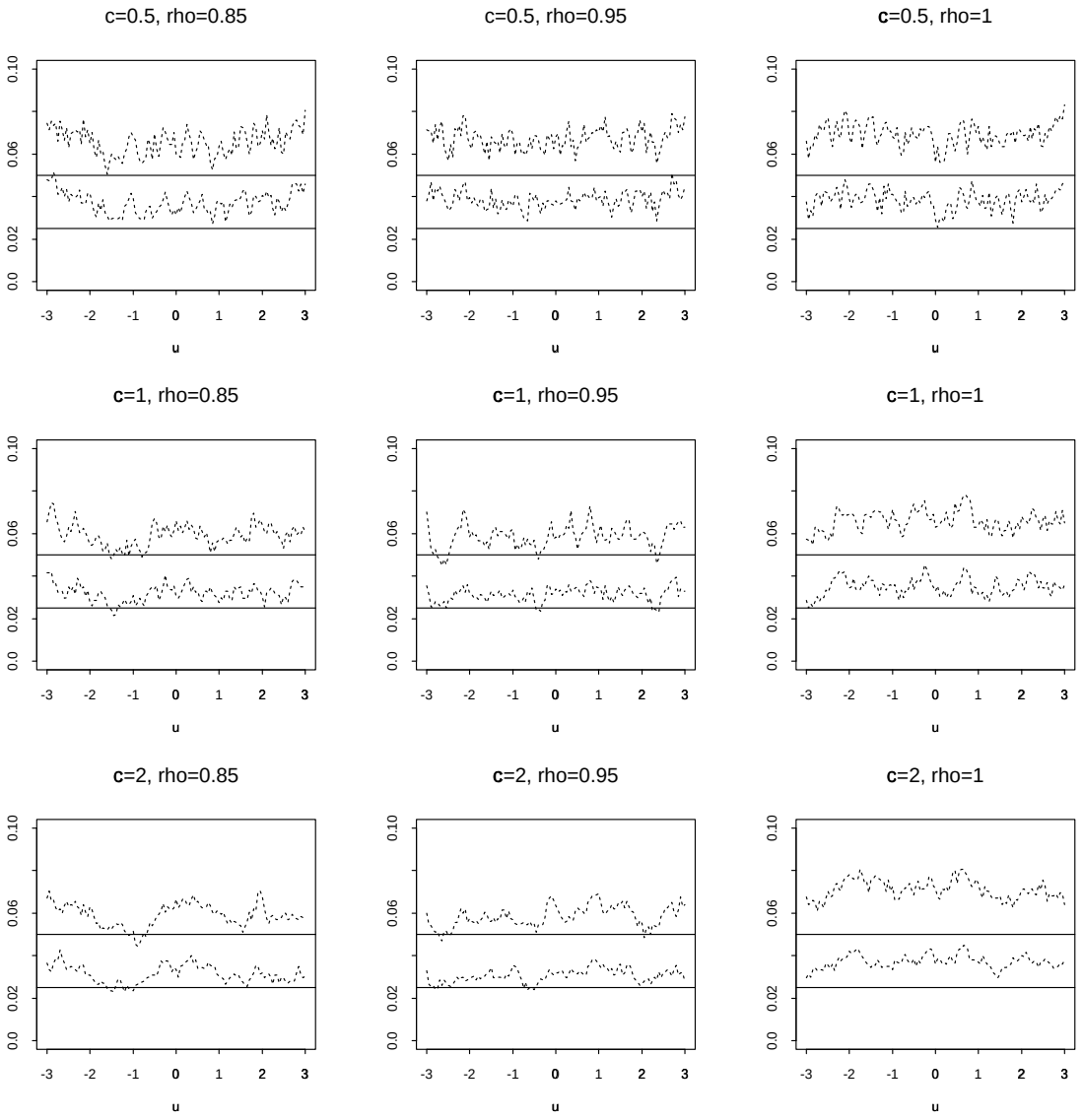
c=0.5, rho=0.85
c=0.5, rho=0.95
c=0.5, rho=1
c=1, rho=0.85
c=1, rho=0.95
c=1, rho=1
c=2, rho=0.85
c=2, rho=0.95
c=2, rho=1
0.0
0.02
0.06
0.10
-3
-2
-1
0
1
2
3
u

Figure 1

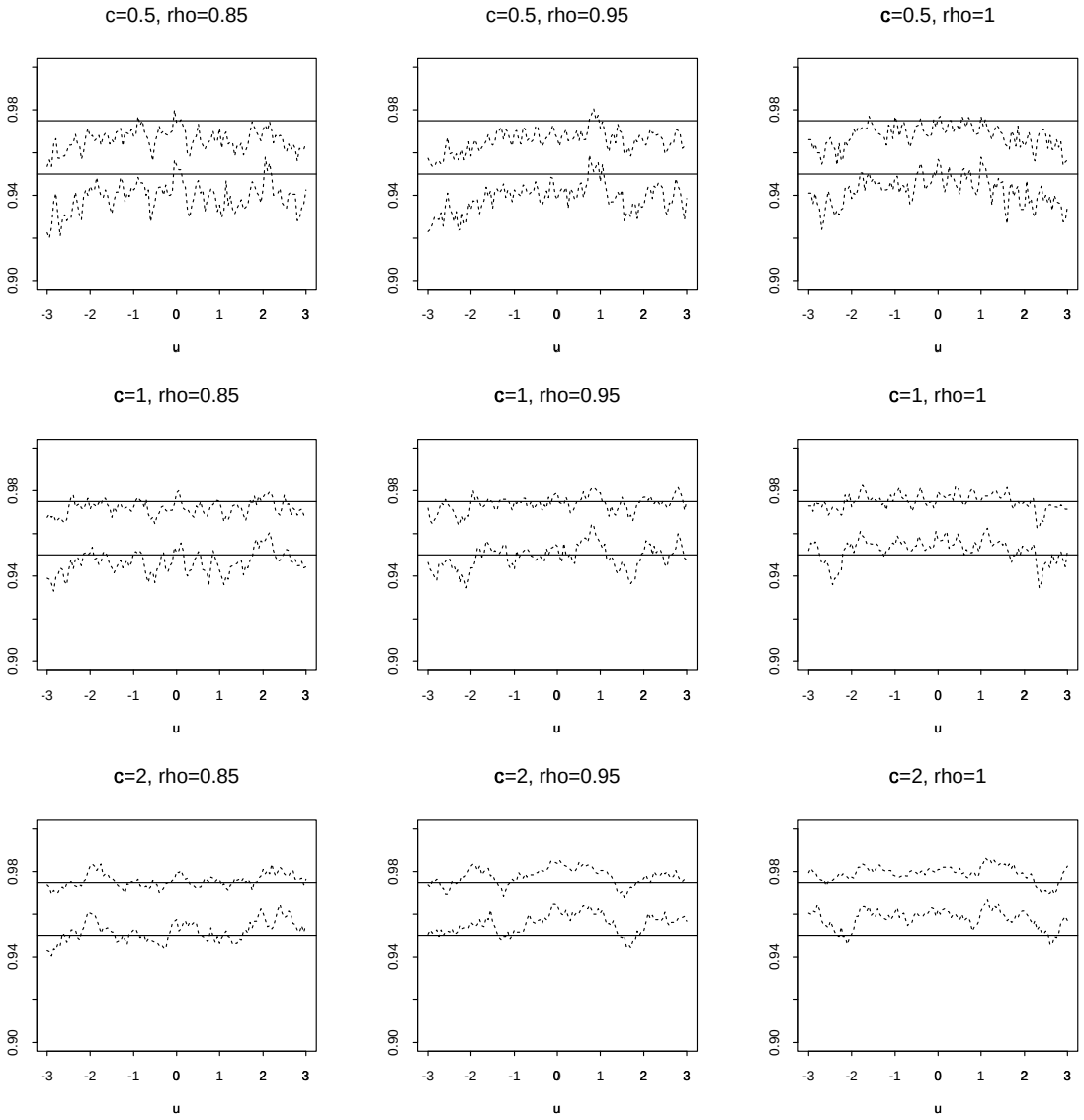
c=0.5, rho=0.85
c=0.5, rho=0.95
c=0.5, rho=1
c=1, rho=0.85
c=1, rho=0.95
c=1, rho=1
c=2, rho=0.85
c=2, rho=0.95
c=2, rho=1
0.90
0.94
0.98
-3 -2 -1 0 1 2 3
u

FIGURE 2

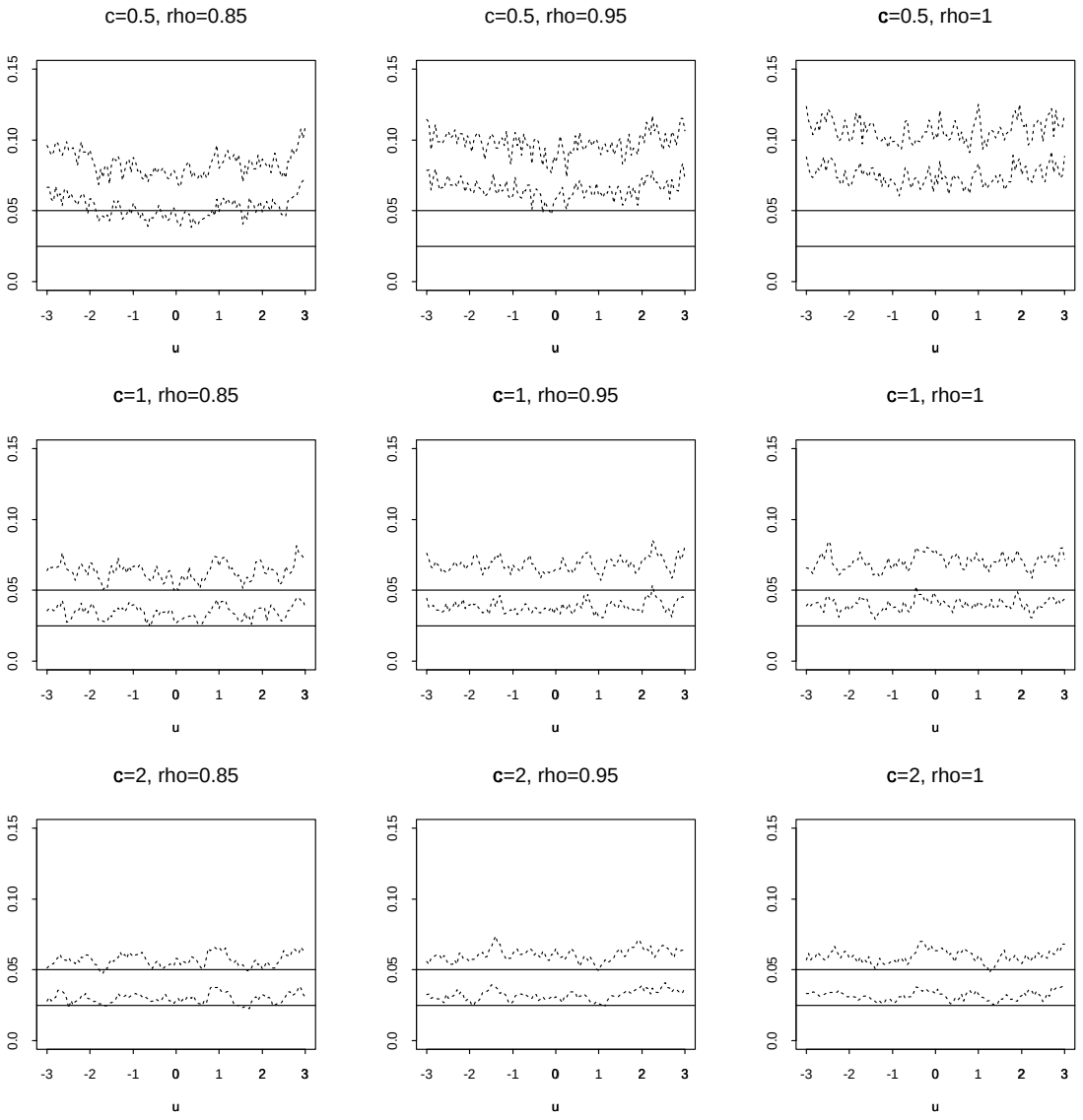
c=0.5, rho=0.85
c=0.5, rho=0.95
c=0.5, rho=1
c=1, rho=0.85
c=1, rho=0.95
c=1, rho=1
c=2, rho=0.85
c=2, rho=0.95
c=2, rho=1
0.0
0.05
0.10
0.15
-3
-2
-1
0
1
2
3
u

Figure 3

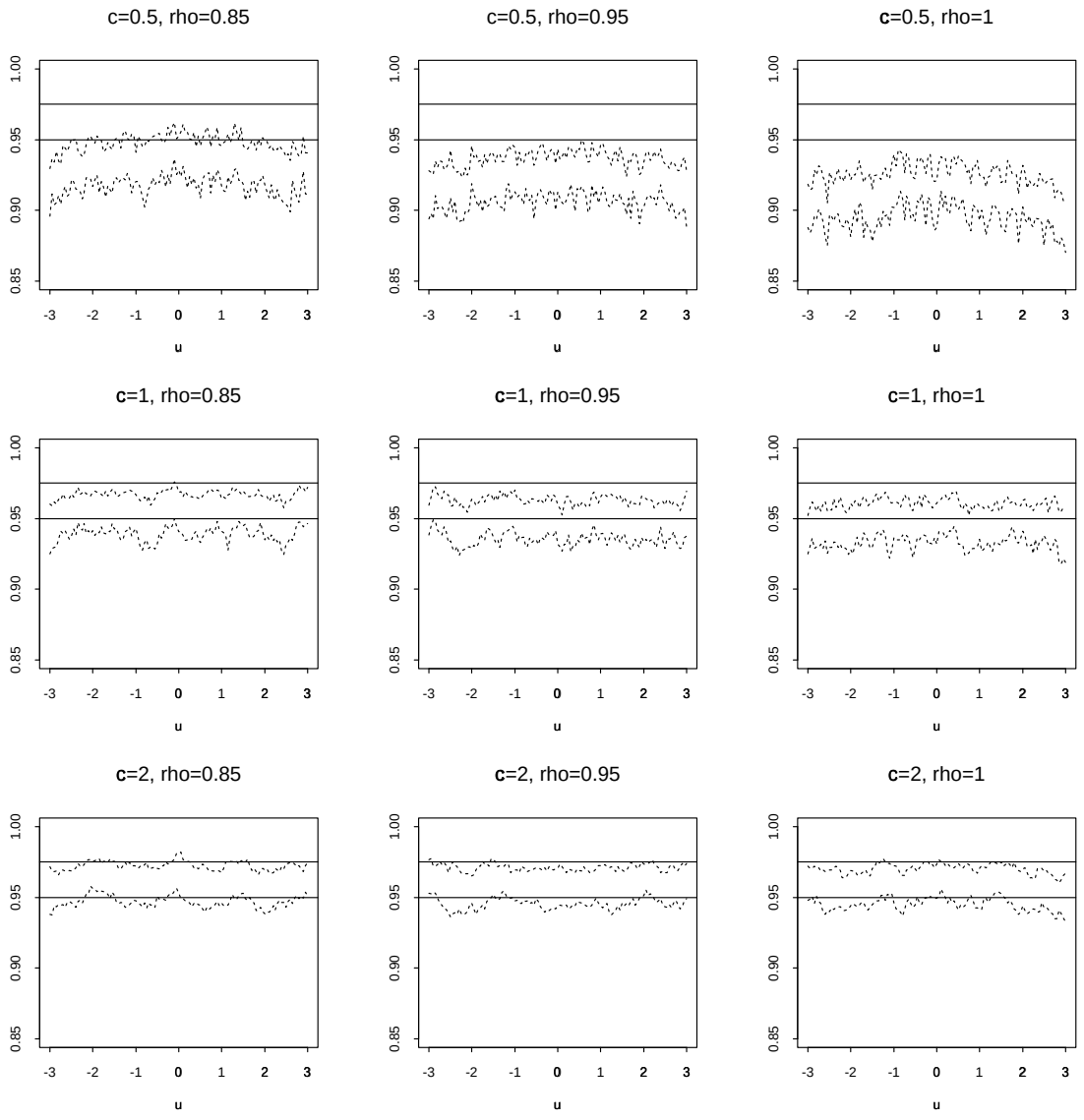
c=0.5, rho=0.85
c=0.5, rho=0.95
c=0.5, rho=1
c=1, rho=0.85
c=1, rho=0.95
c=1, rho=1
c=2, rho=0.85
c=2, rho=0.95
c=2, rho=1
1.00
0.95
0.90
0.85
-3 -2 -1 0 1 2 3
u

FIGURE 4

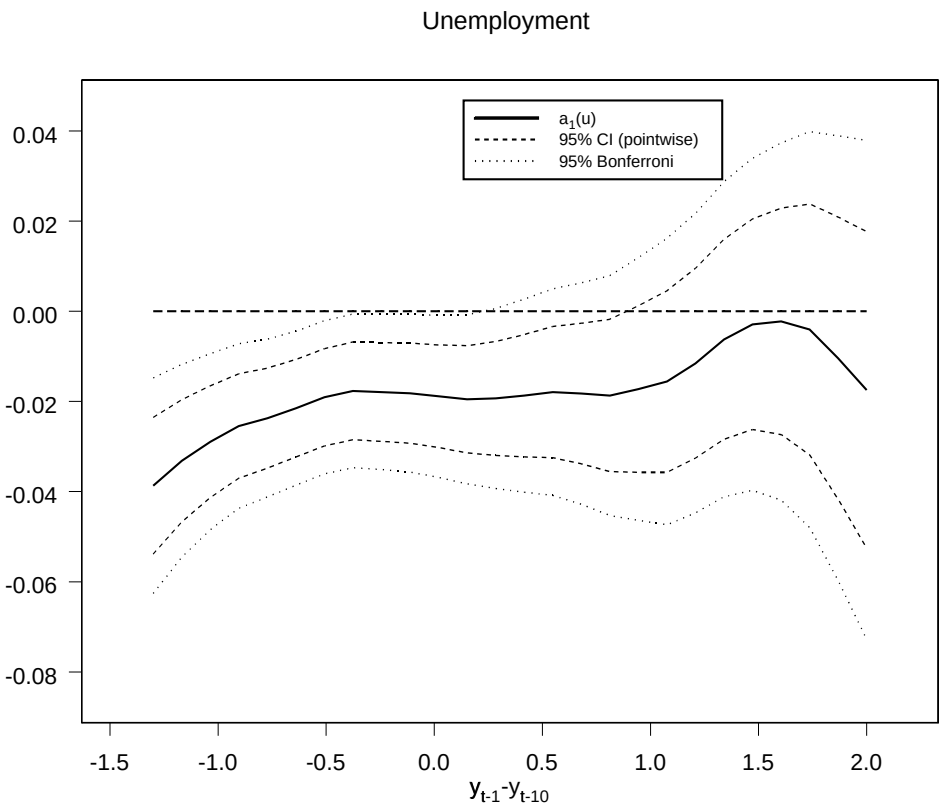
Unemployment
a1(u)
95% CI (pointwise)
95% Bonferroni
0.04
0.02
0.00
-0.02
-0.04
-0.06
-0.08
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
2.0
yt-1-yt-10

Figure 5

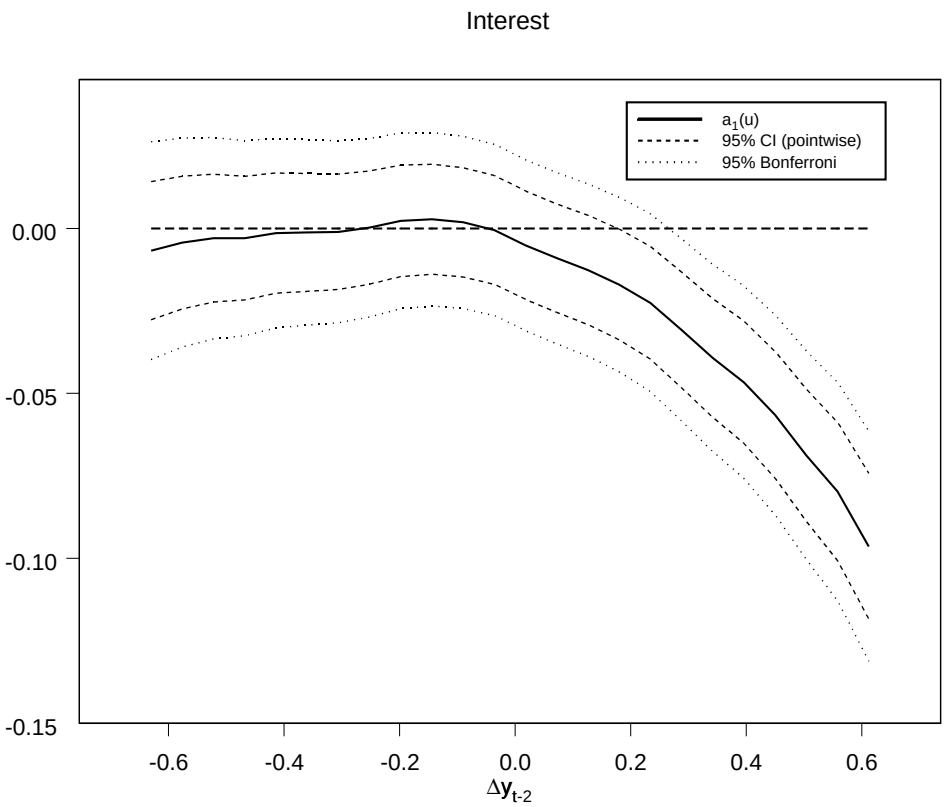
Interest
a1(u)
95% CI (pointwise)
95% Bonferroni
0.00
-0.05
-0.10
-0.15
-0.6
-0.4
-0.2
0.0
0.2
0.4
0.6
Δyt-2

FIGURE 6

References

Bacon, D. W., and D. G. Watts (1971): "Estimating the Transition Between Two Intersecting Lines," *Biometrika*, 58, 525–534.

Cai, Z., J. Fan, and Q. Yao (2000): "Functional-Coefficient Regression Models for Nonlinear Time Series," *Journal of the American Statistical Association*, 95, 941–956.

Caner, M., and B. E. Hansen (2001): "Threshold Autoregression with a Unit Root," *Econometrica*, 69, 1555–1596.

Chan, K. S. (1991): "Percentage Points of Likelihood Ratio Tests for Threshold Autoregressions," *Journal of the Royal Statistical Society, Series B*, 53, 691–696.

——— (1993): "Consistency and Limiting Distribution of the Least Squares Estimator of a Threshold Autoregressive Model," *The Annals of Statistics*, 21, 520–533.

Chan, K. S., and R. S. Tsay (1998): "Limiting Properties of the Least Squares Estimator of a Continuous Threshold Autoregressive Model," *Biometrika*, 45, 413–426.

Chen, R., and R. S. Tsay (1993): "Functional-Coefficient Autoregressive Models," *Journal of the American Statistical Association*, 88, 298–308.

Cox, D. D., and I. Llatas (1991): "Maximum Likelihood Type Estimation for Nearly Nonstationary Autoregressive Time Series," *The Annals of Statistics*, 19, 1109–1128.

Davidson, J. (1994): *Stochastic Limit Theory*. Oxford: Oxford University Press.

Dickey, D. A., and W. A. Fuller (1979): "Distribution of the Estimators for Autoregressive Time Series with a Unit Root," *Journal of the American Statistical Society*, 74, 427–431.

Fan, Y., and Q. Li (1999): "Root-N-Consistent Estimation of Partially Linear Time Series Models," *Journal of Nonparametric Statistics*, 11, 251–269.

Hall, P., and C. Heyde (1980): *Martingale Limit Theory and its Application*. New York: Academic Press.

Hansen, B. E. (1992): "Convergence to Stochastic Integrals for Dependent Heterogeneous Processes," *Econometric Theory*, 8, 489–500.

——— (1995): "Rethinking the Univariate Approach to Unit Root Testing," *Econometric Theory*, 11, 1148–1171.

——— (1996): "Inference when a Nuisance Parameter is not Identified Under the Null Hypothesis," *Econometrica*, 64, 413–430.

——— (1997): "Inference in TAR Models," *Studies in Nonlinear Dynamics and Econometrics*, 1, 119–131.

HASTIE, T. J., AND R. J. TIBSHIRANI (1993): "Varying-Coefficient Models," *Journal of the Royal Statistical Society, Series B*, 55, 757–796.

HÄRDLE, W. (1989): "Asymptotic Maximal Deviation of M-Smoothers," *Journal of Multivariate Analysis*, 29, 163–179.

KURTZ, T., AND P. PROTTER (1991): "Weak Limit Theorems for Stochastic Integrals and Stochastic Differential Equations," *The Annals of Probability*, 19, 1035–1070.

LI, Q. (1999): "Consistent Model Specification Tests for Time Series Econometric Models," *Journal of Econometrics*, 92, 101–148.

LUCAS, A. (1995): "Unit Root Tests Based on M Estimators," *Econometric Theory*, 11, 331–346.

MOKKADEM, A. (1988): "Mixing Properties of ARMA Processes," *Stochastic Processes and Their Application*, 29, 309–315.

MONTGOMERY, A. L., V. ZARNOWITZ, R. S. TSAY, AND G. C. TIAO (1998): "Forecasting the U.S. Unemployment Rate," *Journal of the American Statistical Association*, 93, 478–493.

NUNES, L. C., C. M. KUAN, AND P. NEWBOLD (1995): "Spurious Break," *Econometric Theory*, 11, 736–749.

PERRON, P. (1989): "The Great Crash, the Oil Price Shock, and the Unit Root Hypothesis," *Econometrica*, 57, 1361–1401.

PHILLIPS, P. C. B., AND J. Y. PARK (1998): "Nonstationary Density Estimation and Kernel Autoregression," Yale University.

PHILLIPS, P. C. B., AND V. SOLO (1992): "Asymptotics for Linear Processes," *The Annals of Statistics*, 20, 971–1001.

SEO, B. (1999): "Distribution Theory for Unit Root Tests with Conditional Heteroskedasticity," *Journal of Econometrics*, 91, 113–144.

TERÄSVIRTA, T. (1994): "Specification, Estimation, and Evaluation of Smooth Transition Autoregressive Models," *Journal of the American Statistical Association*, 89, 208–218.

TONG, H. (1978): "On a Threshold Model," in *Pattern Recognition and Signal Processing*, ed. by C. H. Chen. Amsterdam: Sijhoff and Noordhoff.

——— (1990): *Nonlinear Time Series: A Dynamical System Approach*. New York: Oxford University Press.

YOSHIHARA, K. (1976): "Limiting Behavior of U-Statistics for Stationary, Absolutely Regular Processes," *Zeitschrift fur Wahrsheinlichkeitstheorie und verwandte Gebiete*, 35, 237–252.